

Code-Layout, Readability and Reusability

Date	15 March 2026
Team ID	SWTID-2026-5339
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation maintained throughout the project.	Yes	Python PEP-8 coding standards followed.
2	Proper File Structure	Files and folders are logically organized.	Yes	Separate modules for Q&A, Quiz, Summary, Explanation, Learning Path, Templates, and Static files.
3	Meaningful Variable Names	Variables clearly represent their purpose.	Yes	Descriptive variable names improve readability.
4	Function / Method Names	Functions are descriptively named.	Yes	Functions such as <code>generate_quiz()</code> , <code>summarize_text()</code> , <code>answer_question()</code> , and <code>recommend_learning_path()</code> are self-explanatory.
5	Code Comments	Inline and block comments explain important logic.	Yes	Comments added for API routes, Gemini prompts, JSON parsing, and exception handling.
6	Modular Design	Code divided into reusable modules.	Yes	Independent modules created for each AI functionality.
7	No Redundant Code	Duplicate or unused code removed.	Yes	Common logic reused across API endpoints.
8	Error Handling	Exceptions handled gracefully.	Yes	FastAPI exception handling and JSON validation implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Question Answering Module	Python, Google Gemini API	/qa API endpoint and Frontend	High
2	Explanation Module	Python, Google Gemini API	/explain endpoint	High
3	Summary Module	Python, Google Gemini API	/summarize endpoint	High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
4	Quiz Generation Module	Python, Google Gemini API	/quiz endpoint	High
5	Learning Path Recommendation Module	Python, Google Gemini API	/learn/recommendations endpoint	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized modular architecture with separate backend, frontend, and AI modules.
Readability	5	Clean code with meaningful naming conventions and proper formatting.
Reusability	5	Independent modules can be reused in other AI-powered educational applications.
Documentation / Comments	5	Code includes sufficient comments and API documentation for maintainability.
Overall Score	5/5	The project follows best coding practices, modular design, proper documentation, and reusable architecture.